

PIR

DETECT

MOTION

What does it do?

A PIR spots movement nearby, such as somebody walking past.

PIR stands for passive infrared. It watches the warmth that bodies give off and reports when that warmth moves, so it notices people and animals rather than furniture.

Your projects can use it to wake something up the moment somebody arrives, and go quiet again when they leave.

Uses

Burglar alarm

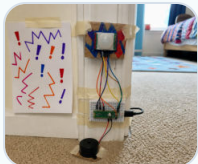


DETECT
Movement in
the room

DECIDE
Nobody
should be in

DO
Raise the
alarm

Sibling alarm



DETECT
Movement at
the door

DECIDE
Someone is
coming

DO
Sound the
warning



Dome
Wide-
angle lens

VCC
Power, 5V

OUT
High on
movement

GND
Ground

Code

● Raspberry Pi

```
from gpiozero import MotionSensor  
  
pir = MotionSensor(4)  
  
moved = pir.motion_detected
```

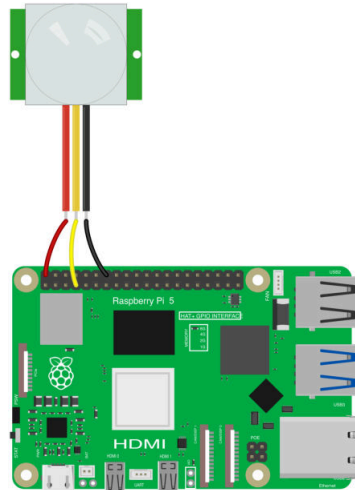
● Raspberry Pi Pico

```
from picozero import MotionSensor  
  
pir = MotionSensor(16)  
  
moved = pir.motion_detected
```

`moved` is True or False. Also `pir.when_motion` and `pir.when_no_motion` to run a function the moment it changes. Most boards carry two dials that set how sensitive it is and how long OUT stays high.

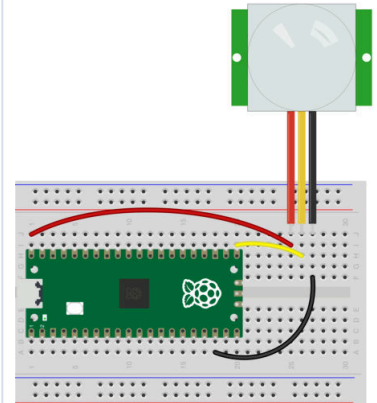
Wiring

RASPBERRY PI 5



5V · OUT GPIO 4 · GND

PICO



VBUS · OUT GP16 · GND